

Python Interview Questions

Python Basic Questions

1. What is **python**?
 1. **Features** of Python?
 2. Python **Frameworks**?
2. Difference between Python 2 and Python 3?
3. What is an **Interpreted language**?
4. What is a **dynamically typed language**?
5. Is **indentation** required in python?
6. ★ What is **PEP 8**?
7. File Extensions in Python?
8. Diff between **.py** and **.pyc** files?
9. How to **get Id of any object**?
10. a=1, b=1 does both have same Id or not?
11. What are modules and packages in Python?

Python Comments Questions

1. Python **Comments**?
2. What is **Docstrings**?
 1. how to **get docstring** in python?
3. Difference **between** Python **comments** and **docstrings**?

Python Variables Questions

1. What is python Variables?
2. Single Quotes or Double Quotes?
3. what is **Global Variables**?
4. what is **global Keyword**?
5. What is **None** in Python?
6. Difference between **None**, **NaN**, and **Null**?
7. How is **memory managed** in Python?
8. What is **Scope** in Python?

Python Operator Questions

1. What is the operator?
2. What is **membership** operators?
3. What is **Identity** operators?
4. Difference between '**==**' and '**is**' Operator?
5. What is **ternary operator** in Python?

Python Data Types Questions

1. What are the common built-in data types in Python?
2. How We can get the data type of any object?
3. **Primitive/Non-Primitive** Data Structures?
4. Type Casting/Type Conversion?
5. Types Of Casting/Type Conversion?

Python conditions Questions?

1. What is If Else?

Python loops questions?

1. Python **Range** Function?
2. Difference between **range()** and **xrange()**?
3. What is the purpose of the range() function?
4. Python **While** Loops?
5. **Switch** Statements?
 1. Switch/Match Statements Using class?
 2. Switch/Match Statements Using function?

Copy

1. what is Copy?
2. ★ What is **deep copy**?
3. ★ what is **shallow copy**?

Other Questions

1. What is an **Iterable**?
2. What is **Iterators**?
3. Diff between **Iterators** and **iterable**?
4. Difference between **generator** and **iterators** in python?
5. What is **closure** Python?
6. what is **context manager**?
7. ★ what is **decorators**?
8. What is a **function**?
9. Difference between arguments and parameters?
10. what is **generator function**?
11. What is **Lambda/Anonymous** Function?
12. ★ What is **GIL(Global Interpreter Lock)**?
13. What is **monkey patching**?
14. ★ What is **Multithreading**?
15. what is **pickling and unpickling**?
16. Python **Json**?
17. single asterisk and double asterick?
18. ***args**(Non-Keyword Arguments) and ****kwargs**(Keyword Arguments)

19. Why use *args and **kwargs in Python?
20. Difference between *args and **kwargs in Python?
21. [Arbitrary Positional Arguments?](#)
22. [Arbitrary Keyword Arguments?](#)
23. [What is MRO\(Method Resolution Order\) / Diamond Problem?](#)
24. [Diamond Problem](#)
25. How does Python resolve methods in Multiple Inheritance?
26. [What is static method?](#)
27. What is **str** and **repr**?
28. [What is the super\(\) Function?](#)
 1. [super\(\) in Multiple Inheritance?](#)
29. [What is **Multithreading**?](#)
30. [How does Python achieve Multithreading?](#)
31. [multithreading vs multiprocessing?](#)
32. [What is I/O-bound Tasks?](#)
33. [What is CPU-bound Tasks?](#)
34. [What is Magic Method Or Dunder Methods?](#)
35. [What is Math Module?](#)
36. [floor\(\) and ceil\(\) Functions?](#)
37. [Python logging?](#)
38. [What is slicing in Python?](#)

Exception Handling

1. [What is Exception Handling?](#)
2. [try, except, else, finally](#)
3. [How can we handle errors in python?](#)
4. [How many ways exception in python?](#)

File Handling

1. [What is File Handling?](#)
2. [What is with?](#)
3. [How do you open and close a file in Python?](#)

Date and Time

1. [DateTime Function](#)

Python String Questions

1. [Python Strings](#)
2. [find the length/Count number of characters in a string](#)
3. [String Concatenation](#)
4. Assign String to a Variable
5. Multiline Strings
6. Check String
7. Slicing Strings
8. Strings Method OR Change/Modify String
9. What is an f-string and how do you use it
10. Format Strings
11. Strings Method
12. capitalize(), count, Upper, Lower, title, replace, strip, lstrip, rstrip split, join, startswith, endswith
13. isalpha, isdigit, isalnum
14. When would you use rfind()
15. d
16. Escape Characters
17. How would you confirm that 2 strings have the same identity
- 18.
19. |

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| No. | Logical string Questions
|
| :---: | -----
|-----|
|      | [Looping Through a String](./1_python/6.0_string.md#ques-Looping-Through-a-String)
|      | [Extract numbers from string](./1_python/6.0_string.md#ques-Extract-numbers-from-string)
|      | [Reversed the String](./1_python/6.0_string.md#ques-Reversed-the-String)
|      | [Reversed word in string?](./1_python/6.0_string.md#reversed-word-in-string)
|      | [Check for Palindrome](./1_python/6.0_string.md#check-for-palindrome)
|      | [Remove vowels from a string](./1_python/6.0_string.md#ques-remove-vowels-from-a-string)
|      | [Find repeated characters in a string python](./1_python/6.0_string.md#ques-Find-repeated-characters-in-a-string-python)
|      | [program to count the frequency of each character](./1_python/6.0_string.md#ques-count-the-frequency-of-each-character)
|      | [Count Vowels and Consonants?](./1_python/6.0_string.md#count-vowels-and-consonants)
|      | [Remove duplicate characters from a string](./1_python/6.0_string.md#remove-duplicate-characters-from-a-string)
|
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Python List Questions

1. What is **List**?
2. List **length**? :- `print(len(list1))`
3. ★ List **comprehension**?
4. **Access** List Items/List Slicing?
5. How to check if an item exists in a list?
6. **Add List** Items?
 1. Append method? :- `list1.append("a")`
 2. Insert Method? :- `list1.insert(1, "apple")`
 3. Extend Method? :- `list1.extend(sets)`
 4. ★ Difference between `append()` and `extend()`?
7. Change/Update List Items?
8. Copy List?
 1. Copy Method?
 2. List Method?
9. Join List?
 1. Plus(+) Operator
 2. append method
 3. extend method
10. **Remove List Item**?
11. `remove()` method?
12. `pop()` Method?
13. `del` keyword?
14. `clear()` method?
15. ★ Difference between `remove()` and `pop()`?
16. **Sort list**?
 1. `thislist.sort()` :- A, B, C, a, b, c
 2. `thislist.sort(key = str.lower)` :- a, b, c, A, B, C
 3. `thislist.sort(reverse = True)` :- c, b, a, C, B, A
 4. `thislist.reverse()` :- a, B, c, D, e
 5. ★ Difference between `sort()` and `sorted()`?
 6. ★ How to reverse a list? :- `list1.reverse()`
17. **Loop List**
18. ★ Flatten list?
19. List Method?
 1. `append()` Method
 2. `extend()` Method
 3. `insert()` Method
 4. `Copy()` Method
 5. `count()` Method
 6. `index()` Method
 7. `pop()` Method
 8. `remove()` Method
 9. `clear()` Method
 10. `reverse()` Method

11. sort() Method

- 20. ★ List vs Tuple
- 21. What is the difference between an array and a list?
- 22. ★ Mutable vs Immutable Objects
- 23. What is Map, Filter, Reduce?

Python List Logical Questions

1. How to find count of the list?
2. **find duplicate elements** in a list?
3. How to **remove duplicates** from a list?
4. Find **Maximum** and **Minimum** Element?
5. Find Second Largest Element?
6. Common Coding Questions on Lists

Reverse a list without built-in functions.

Find largest/smallest element.

Rotate a list.

Check if a list is palindrome.

Find intersection of two lists.

Find common elements in two lists.

Move all zeros to the end.

Flatten a nested list.

Find frequency of each element.

Split a list into chunks.

Sort a list without using sort().

★ Most Frequently Asked in Interviews

List vs Tuple

List Comprehension

append() vs extend()

remove() vs pop()

sort() vs sorted()

Shallow Copy vs Deep Copy

List Slicing

Flatten List

Remove Duplicates

Reverse List

Nested List

Time Complexity of List Operations

| [Find the Length of a List?]

(./2_list/list_logical_ques/Length_of_a_List.md#find-the-length-of-a-list)

|

| [Length hint function?]

(./2_list/list_logical_ques/Interchange_first_and_last_elements_in_a_list.md#ques-interchange-first-and-last-elements-in-a-list) |

| [Finding length of list using loop?]

(./2_list/2.10_List_questions.md#find-the-length-of-a-list)

|

| [Interchange first and last elements in a list?]

(./2_list/2.10_List_questions.md#interchange-first-and-last-elements-in-a-list)

|

| [Swap Two Elements in a List?](./2_list/2.10_List_questions.md#swap-two-elements-in-a-list) |

| [Swap elements in String list?](./2_list/2.10_List_questions.md#swap-elements-in-string-list)

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|      | [find the single number(unique-element) of the list?](#ques-find-the-
single-numberunique-element-of-the-list)
|
|      | [find the even number from the list?](#ques-find-the-even-number-from-
the-list)
|
|      | [Print duplicate list, Find Even Or Odd Number?](#ques-print-duplicate-
list-find-even-or-odd-number)
|
|      | [Find the duplicate element from list?](#find-the-duplicate-element-
from-list)
|
|      | [Remove duplicate item from list using List comprehension?]
(./2_list/list_logical_ques/Remove_the_negative_index_from_the_list.md)
|
|      | [Convert a list into string?](#ques-convert-a-list-into-string)
|
|      | [Write a program to print a list in reverse order?](#ques-write-a-
program-to-print-a-list-in-reverse-order)
|
|      | [find the max, min number from the list user input?](#ques-find-the-max-
min-number-from-the-list-user-input)
|
|      | [find the sum of list elements?](#ques-find-the-sum-of-list-elements)
|
|      | [Generate a number list between two ranges?](#ques-generate-a-number-
list-between-two-ranges)
|
|      | [Remove elements in a list after a specific index?](#ques-remove-
elements-in-a-list-after-a-specific-index)
|
|      | [Remove elements in a list before a specific index?](#ques-remove-
elements-in-a-list-before--a-specific-index)
|
|      | [Remove elements in a list between 2 indices?](#ques-remove-elements-in-
a-list-between-2-indices)
|
|      | [How would you convert a list to an array?](#ques-how-would-you-convert-
a-list-to-an-array)
|

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Python Tuple Questions

- 📌 Same as list, firstly convert into list then again into tuple.

1. What is **Tuple**?
2. Create Tuple With One Item?
3. Tuple **length**?
4. Check if Item Exists in tuple?
5. Tuple() Constructor?
6. Access tuple items
7. Add Tuple Items
 1. Append method?
 2. Insert Method?
 3. Extend Method?
8. Change/Update items
9. copy Items
10. Join Tuple
11. Remove items
12. Sort Item
13. Loop Item
14. Tuple constructor
15. **Packing and Unpacking**
16. Tuple Methods
17. ★ Difference between List and Tuple?

Python Set Questions

1. What is Set?
2. Length of a Set?
3. set() Constructor?
4. Add Items of set?
5. Access Items of set?
6. Update Method?
7. Remove Item?
8. Loop Sets?
9. Join Set?
10. Set Methods?
11. add() Methods
12. copy()
13. difference()
14. difference_update()
15. isdisjoint()
16. issubset()
17. issuperset()
18. remove()
19. discard()
20. pop()
21. clear()
22. union()
23. update()
24. intersection()
25. intersection_update()
26. symmetric_difference()
27. symmetric_difference_update()
28. What is difference between Discard() and Remove()?

Python Dictionary Questions

1. What is Dictionary?
2. Dictionary Length?
3. dict() Constructor?
4. How to Check if Item Exists?
5. Access item of the Dictionary?
6. Add item in a Dictionary?
7. Change/Update in a Dictionary?
 1. Using Key name
 2. using update method
8. Copy Dictionary?
9. Remove Dictionary?
10. nested Dictionary?
11. loop Dictionary?
12. Dictionary Method?

Python Dictionary Logical Questions

1. How to Merging Or Adding two Dictionaries?
2. Combine two dictionary adding values for common keys?
3. What will the output of following Program?
4. Sort in dictionary with key, Value and Items
5. Can we use tuple as keys inside python dictionary?
6. Can we use lists as keys inside python dictionary?
7. What is enumerate function inside a dictionary?
8. How can combine two dictionaries together?
9. Find Unique dictionary from list by id?

Python OOPS

1. What is Object-Oriented Programming (OOPS)?
2. What is **break**, **continue** and **pass** in Python?
3. ★ Difference between **method** and **function**?
4. What is Python Access Modifiers?
 1. Public Member
 2. Protected Member
 3. Private Member
5. what is Name **Mangling** in Python?
6. What is Method?
7. What is Method Overloading?
8. What is Method Overriding?

Class in oops

1. What is **Class**?
 1. Define a class?
 2. Get Class Variables/attributes?
 1. Using Class Name
 2. using **getattr()** function
 3. Using Object
 3. Change values for class variables?
 1. Using class
 2. using **setattr()** function
 3. Using **Object**
 4. **Delete** class variables?
 1. using **delattr()** function:
 2. using **del** keyword
2. Built-In Class Functions?
 1. getattr() Function
 2. setattr() function
 3. delattr() function
 4. hasattr() function
3. What is **issubclass()**?
4. Built-In Class Attributes?
5. ★ what is **self** keyword?
6. What is **init** Method?
7. Difference between **Class Variable** and **Instance Variable**?
8. What is an Instance Method?
9. What is a Class Method?
10. What is **cls** keyword?
11. What is **slots** in Python?
12. What is Concrete Method?
13. What is difference between **repr()** and **str()**?

Objects in oops

1. What is **Object**?
2. Delete the Object?
3. Delete the properties of the Object?
4. Counting the Number of objects of a Class?

Encapsulation

1. What is Encapsulation?
2. Getter and Setter Methods?

Polymorphism

1. What is Polymorphism?
2. Types of Polymorphism?
 1. Compile time polymorphism (Static) - Method Overloading
 2. Runtime time polymorphism (Dynamic) - Method Overriding

Inheritance

1. What is Inheritance?
2. Why do we use Inheritance?
3. Types Of Inheritance?
 1. Single Inheritance
 2. Multiple Inheritance
 3. Multi-Level Inheritance
 4. Hierarchical Inheritance
 5. Hybrid Inheritance

constructor and Destructors

1. what is constructor?
2. Types of Constructors?
3. Constructor Overloading?
4. Constructor With Default Values?
5. What is Destructors?

Abstract Class and interface

1. What is Abstract Class?
 1. When use abstratc class?
 2. How to Create an Abstract Method?
2. What is Interface?
3. Abstract Class vs Interface?

Basic Programs

1. Print Hello World
2. Swap Two variables
3. **Even or odd**:- check number

4. Even or odd:- without any condition?
5. Even or odd:- Print number
6. **Prime Number**:- Check Number?
7. Prime Number:- Print Number?
8. ★ Find the **Factorial** of a Number?
9. **Reverse**:- string?
10. Reverse:- reverse a word sentence?
11. Reverse:- Number?
12. ★ Check number is an **Armstrong number** or not?
13. ★ Two given **words are anagrams**?
14. ★ Find **Fibonacci** series up to n
15. ★ String is a **Palindrome**?
16. ★ Number is a **Palindrome**?
17. list Sorting using **bubble sort**?
18. **FizzBuzz** problem?

List Programs

1. ★ Find **Largest Element** in an Array/list?
2. ★ Find the Duplicate list and Unique list?
3. Find sum of array/list?
4. add one digit?
5. convert list to dictionary?
6. convert a list to string?

String Programs

1. Find the first latter of the Word?
2. Find the **vowels** in a string?

- ✓ init()
- ✓ Class Variables vs Instance Variables
- ✓ Instance Methods
- ✓ Built-in Class Functions
- ✓ Built-in Class Attributes

+ Add:

Difference between Class Variable and Instance Variable?

What is an Instance Method?

What is a Class Method?

What is cls keyword?

💡 Encapsulation

- ✓ What is Encapsulation?

+ Add:

Public, Protected and Private Members?

Getter and Setter Methods?

🔴 Inheritance

+ Add:

Why do we use Inheritance?

How to call Parent Class Constructor?

Difference between Inheritance and Composition?

🔴 Polymorphism

✅ Method Overloading

✅ Method Overriding

+ Add:

Does Python support Method Overloading?

Operator Overloading?

Duck Typing?

🔴 Abstraction

✅ Abstract Class

✅ Interface

+ Add:

Why can't we create objects of Abstract Class?

abc module?

@abstractmethod decorator?

🔴 Constructor & Destructor

✅ Constructor

✅ Types of Constructors

✅ Destructor

+ Add:

Constructor Chaining?

Can a Constructor return a value?

🔴 Special Methods (Magic/Dunder Methods)

✅ str()

✅ repr()

+ Add:

len()

eq()

add()

call()

🔴 Class Methods & Static Methods

✅ Static Method

+ Add:

What is @classmethod?

Difference between Instance Method, Class Method and Static Method?

Interview Favorite Question:

Method Type First Parameter

Instance Method self

Class Method cls

Static Method No self/cls

+ Add:

How does Python resolve methods in Multiple Inheritance?

What is mro() method?

+ Add:

🗨 Object Lifecycle

+ Add:

Object Creation Process?

Garbage Collection in Python?

Reference Counting?

★ Most Asked Interview Questions

Four Pillars of OOPS?

Difference between Class and Object?

Difference between Encapsulation and Abstraction?

Difference between Method Overloading and Overriding?

Difference between Class Variable and Instance Variable?

Difference between Class Method and Static Method?

What is Abstract Class?

What is Interface?

What is Constructor?

What is Destructor?

What is init()?

What is self?

What is cls?

What is Polymorphism?

What is Inheritance?

What is Encapsulation?

What is Abstraction?

Software enginring Questions

1. What is the difference between authentication and authorization?

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| No. 📄 | [Python Interview Questions](./0.0_python_questions.md)
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| :---: | -----
|-----|
|       | [Date Time Function?](./1_python/DateTime_Function.md#attributes-of-
datetime-module) |
|       | [Debug code?](./1_python/debug_code.md)
|
|       | [Exception Handling]
(./1_python/Exception_and_error_Handling.md#exception-handling) |
|       | [What is Python JSON?](./1_python/1.0_python.md#ques-what-is-python-
json) |
|       | [floor() and ceil() Functions?](./1_python/1.0_python.md#ques-floor-and-
ceil-functions) |

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| No. | Python Tuple Program Interview Questions
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| :---: | -----
|-----|
|       | [Convert a list into a tuple?](./3_tuple/3.9_tuple_questions.md#ques-
convert-a-list-into-a-tuple) |
|       | [How to Sort List Of Tuples By The First Element?]
(./3_tuple/3.9_tuple_questions.md#ques-how-to-sort-list-of-tuples-by-the-first-
element) |
|       | [How to Sort List Of Tuples By The Second Element?]
(./3_tuple/3.9_tuple_questions.md#ques-how-to-sort-list-of-tuples-by-second-
element) |

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| No. | Python Oops Interview Questions
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| :---: | -----
|-----|
|       | [Object-Oriented Programming (OOPS)?](./1_python/7_oops.md#object-
oriented-programming-oops) |
|       | [What is the use of **self** in Python?](./1_python/7_oops.md#ques-what-
is-the-use-of-self-in-python) |
|       | [What is **break**, **continue** and **pass** in Python?]
(./1_python/7_oops.md#ques-what-is-break-continue-and-pass-in-python) |
|       | [What is the super() Function?](./1_python/7_oops.md#ques-what-is-the-
super-function) |
|       | [What is **issubclass**()?](./1_python/7_oops.md#ques-what-is-
issubclass) |
|       | [What is **__str__** and **__repr__**?](./1_python/7_oops.md#ques-
what-is-str-and-repr) |
|       | [What is **Abstract** Method?](./1_python/7_oops.md#ques-what-is-

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abstract-method)
|
| [What is **Concrete** Method?](./1_python/7_oops.md#ques-what-is-
concrete-method)
|
| [Difference between **method** and **function**?]
(./1_python/7_oops.md#difference-between-method-and-function)
|
| [Accessing the attributes value different ways?]
(./1_python/7_oops.md#accessing-the-attributes-value-different-ways)
|
| [Change the value of the attribute?](./1_python/7_oops.md#change-the-
value-of-the-attribute)
|
| [What is __init_ _ Method?](./1_python/7_oops.md#ques--what-is-init-
method)
|
| [Delete the Object?](./1_python/7_oops.md#ques-delete-the-object)
|
| [Python Access Modifiers?](./1_python/7_oops.md#python-access-modifiers)
|
| [What is Python **Enumeration**?](./1_python/7_oops.md#what-is-python-
enumeration)
|
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| [What is class meta?](#what-is-class-meta)
|
| -----
|
| -----
|
| [private attributes and method?](#private-attributes-and-method)
|
| [What is the super() Function](#ques-what-is-the-super-function)
|
| [What is Abstract Class?](#ques-what-is-abstract-class)
|
| [When use abstratc class?](#ques-when-use-abstratc-class)
|
| [How to Create an Abstract Method](#how-to-create-an-abstract-method)
|
| [Built-In Class Attributes?](#built-in-class-attributes)
|
| [What is Method Overloading?]
(./7_oops/Method_Overloading_Overriding.md#ques-what-is-method-overloading)
|
| [What is Method Overriding?]
(./7_oops/Method_Overloading_Overriding.md#ques-what-is-method-overriding)
|
| [What is Method?](#ques-what-is-method)
|
| [What is static method?](./7_oops/static_method.md#what-is-static-
method)
|

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<!-- ### Oops
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1. What is Object?
 1. Modifying Object's properties.
2. What is Polymorphism?
 1. Types of Polymorphism?
 1. Method overloading/ Duck Typing/ Compile time polymorphism (Static)
 2. Method Overriding/ Runtime time polymorphism (Dynamic) -->

Python Programs Questions

No.	Python array Programs Questions
:---	-----
	[Find the missing number in the array?](../7_python_program.md#find-the-missing-number-in-the-array)
No.	Math Module
:---	-----
	[what is Math Module?](../1_python/math_module.md#math-module)
	[math.floor() Method](../1_python/math_module.md#mathfloor-method)
	[math.ceil() Method](../1_python/math_module.md#mathceil-method)

No.	Python Other Questions
:---	-----
	[Arbitrary Arguments](../1_python/args_and_kwargs.md#arbitrary-arguments)

No.	[Questions](../0.0_python_questions.md)
:---	-----
	[floor() and ceil() Functions?](#ques-floor-and-ceil-functions)
	[What is Generator Functions?](#ques-what-is-generator-functions)
	[Positional (non-keyword) arguments](#positional-non-keyword-arguments)
	[Keyword arguments](#keyword-arguments)
	[DateTime Function](#datetime-function)
	[What is File Handling in Python?](#ques-what-is-file-handling-in-python)

Python Interview Questions

4. Explain how can you make a Python Script executable on Unix?
5. What are Python namespaces? Why are they used?
6. What is Scope Resolution in Python?
9. How do you copy an object in Python?
12. What is PYTHONPATH in Python?
13. What is the use of help() and dir() functions?
16. How are arguments passed by value or by reference in python?
17. What are iterators in Python?

18. Explain how to delete a file in Python?
19. Explain split() and join() functions in Python?
20. What are negative indexes and why are they used?

Python OOPS Interview Questions

3. How do you access parent members in the child class?
4. Are access specifiers used in python?
5. Is it possible to call parent class without its instance creation?
6. How is an empty class created in python?
7. Differentiate between new and override modifiers.
8. Why is finalize used?
9. What is init method in python?
10. How will you check if a class is a child of another class?
- 11.

Python Libraries Interview Questions

1. Ques:-What is the Python "with" statement designed for?

Python Collections (Arrays)?

There are four collection data types in the Python programming language:

- * **List** is a collection which is ordered and changeable. Allows duplicate members.

- * **Tuple** is a collection which is ordered and unchangeable. Allows duplicate members.

- * **Set** is a collection which is unordered and unindexed. No duplicate members.

- * **Dictionary** is a collection which is unordered, changeable and indexed. No duplicate members.

**Ques. What is built-in module in Python?*

<https://docs.python.org/3/py-modindex.html>

Example

```
```python
>>> import html
>>> import html.parser
import mysql.connector
```

ye built in package hote hai

agar or karne hai to pip ki help se karenge

String to array:-

```
thislist = ["apple", "banana", "cherry"]
str1 = ' '.join(thislist)
counter = dict.fromkeys(str1, 0)
print(counter)
for item in str1:
 counter[item] += 1
print(counter)
```

```
import collections
print(collections.Counter("hello"))
```

## Use of "get()" function

### What are Python namespaces?

A namespace in python refers to the name which is assigned to each object in python. The objects are variables and functions. As each object is created, its name along with space (the address of the outer function in which the object is), gets created. The namespaces are maintained in python like a dictionary where the key is the namespace and value is the address of the object. There are 4 types of namespace in python-

Built-in namespace- These namespaces contain all the built-in objects in python and are available whenever python is running. Global namespace- These are namespaces for all the objects created at the level of the main program. Enclosing namespaces- These namespaces are at the higher level or outer function. Local namespaces- These namespaces are at the local or inner function.

```
a = "mohit kumar"
new_list = [1,2,3,4,3,2,1,5,6,7]
list4 = []
for item in a:
 list4.append(item)
print(list4)
list5 = []
[list5.append(item*5) for item in new_list if item not in list5]
print(list5)
dict1 = {}
for item in List:
dict1[len(item)] = item
print(dict1)
dict1 = {len(item): item for item in List}
print(dict1)
abc = sorted(dict1)
list6 = [dict1[item] for item in abc]
print(list6)
```

g. WAP to count from a string? l. Count occurrence of each number in list. s. Write a custom insert() function for list eg. def custom\_insert (list, index, item). bb. Can we pass list or tuple as a key in dictionary? ee. What is property decorator? ii. What is lazy evaluation in python? b. Static file contains which type of file normally? c. How can be file read in specific location? d. How do you remove a file from a folder? e. If user upload excel file check file format if it is valid or invalid. g. What is Garbage Collector? g. What is the difference between Static method and Class method? h. Private variable and how we can access that? j. Difference multilevel and multiple inheritance and drawback? a. How can we handle errors in python? b. How many ways exception in python?

1. Numpy a. Tell me some python libraries .. Numpy b. Convert a list into numpy

i. Count occurrence of each number in list. bb. Can we pass list or tuple as a key in dictionary? ee. What is property decorator? ii. What is lazy evaluation in python? b. Static file contains which type of file normally? c. How can be file read in specific location? d. How do you remove a file from a folder? e. If user upload excel file check file format if it is valid or invalid. g. What is Garbage Collector? g. What is the difference between Static method and Class method? h. Private variable and how we can access that? j. Difference multilevel and multiple inheritance and drawback?

<https://pynative.com/python-constructors/>

##### chatgpt Certainly, here are some Python interview questions across different levels of complexity:

Beginner:

How can you iterate over a dictionary?

Intermediate:

Explain the Global Interpreter Lock (GIL) in Python. How does it affect multi-threading? How does exception handling work in Python? Provide examples of using try, except, finally.

Advanced:

What are generators and iterators in Python? How are they different from each other? Explain the concept of "duck typing" in Python. How can you profile and optimize Python code for performance? Explain the concept of metaclasses in Python. What are async functions in Python? How do they relate to asynchronous programming?

Problem-Solving:

Write a function to check if a given string is a palindrome. Implement a binary search algorithm. Given a list of numbers, write a function to find the two numbers that add up to a specific target sum. Implement a basic LRU (Least Recently Used) cache. Remember, interview questions can vary widely based on the specific role and company. It's important to not only memorize answers but also to understand the underlying concepts. Practice coding these questions and explaining your thought process out loud, as interviews often focus on problem-solving skills and how well you can communicate your solutions. #####

AbstractUser vs AbstractBaseUser The default User model in Django uses a username to uniquely identify a user during authentication. If you'd rather use an email address, you'll need to create a custom User model by either subclassing AbstractUser or AbstractBaseUser.

Options:

AbstractUser: Use this option if you are happy with the existing fields on the User model and just want to remove the username field. AbstractBaseUser: Use this option if you want to start from scratch by creating your own, completely new User model.

[https://books.agiliq.com/projects/django-orm-cookbook/en/latest/null\\_vs\\_blank.html](https://books.agiliq.com/projects/django-orm-cookbook/en/latest/null_vs_blank.html)

## Python interview questions

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### Ques. What are global, protected and private attributes in Python?

- **Global variables** are public variables that are defined in the global scope. To use the variable in the global scope inside a function, we use the **global** keyword.
- **Protected attributes** are attributes defined with an **single underscore** prefixed to their identifier eg. `_mohit`. They can still be accessed and modified from outside the class they are defined in but a responsible developer should refrain from doing so.
- **Private attributes** are attributes with **double underscore** prefixed to their identifier eg. `__mohit`. They cannot be accessed or modified from the outside directly and will result in an `AttributeError` if such an attempt is made. [↑ Back to Top](#)

### Ques. How would you convert a list to an array?

- Python list is a linear data structure that can hold heterogeneous elements. Python does not have a built-in array data type. If you want to create an array in Python, then use the numpy library.
- To *install numpy* in your system, type the following command.
- `python3 -m pip install numpy`
- **1. Using `numpy.array()`**

```
import numpy as np

elon_list = [11, 21, 19, 18, 29]
elon_array = np.array(elon_list)

print(elon_array)
print(type(elon_array))
```

Output:-  
[11 21 19 18 29]

```
<class 'numpy.ndarray'>

2. Using numpy.asarray()
import numpy as np

elon_list = [11, 21, 19, 18, 29]
elon_array = np.asarray(elon_list)

print(elon_array)
print(type(elon_array))

Output:- [11 21 19 18 29]
<class 'numpy.ndarray'>
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## Ques. What is Map, Filter, Reduce?

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## Why is NumPy Faster Than Lists?

### 1. Homogeneous Data Types:

- **NumPy** arrays store elements of the **same data type**, when Python **lists** can **hold mixed data types**.

## 2. Broadcasting

- NumPy supports broadcasting, which allows for arithmetic operations between arrays of different shapes. This feature avoids the creation of temporary arrays and reduces computational overhead, making operations more efficient.

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dict1 = {len(item): item for item in List}
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abc = sorted(dict1)
list6 = [dict1[item] for item in abc]
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## Django interview Questions

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